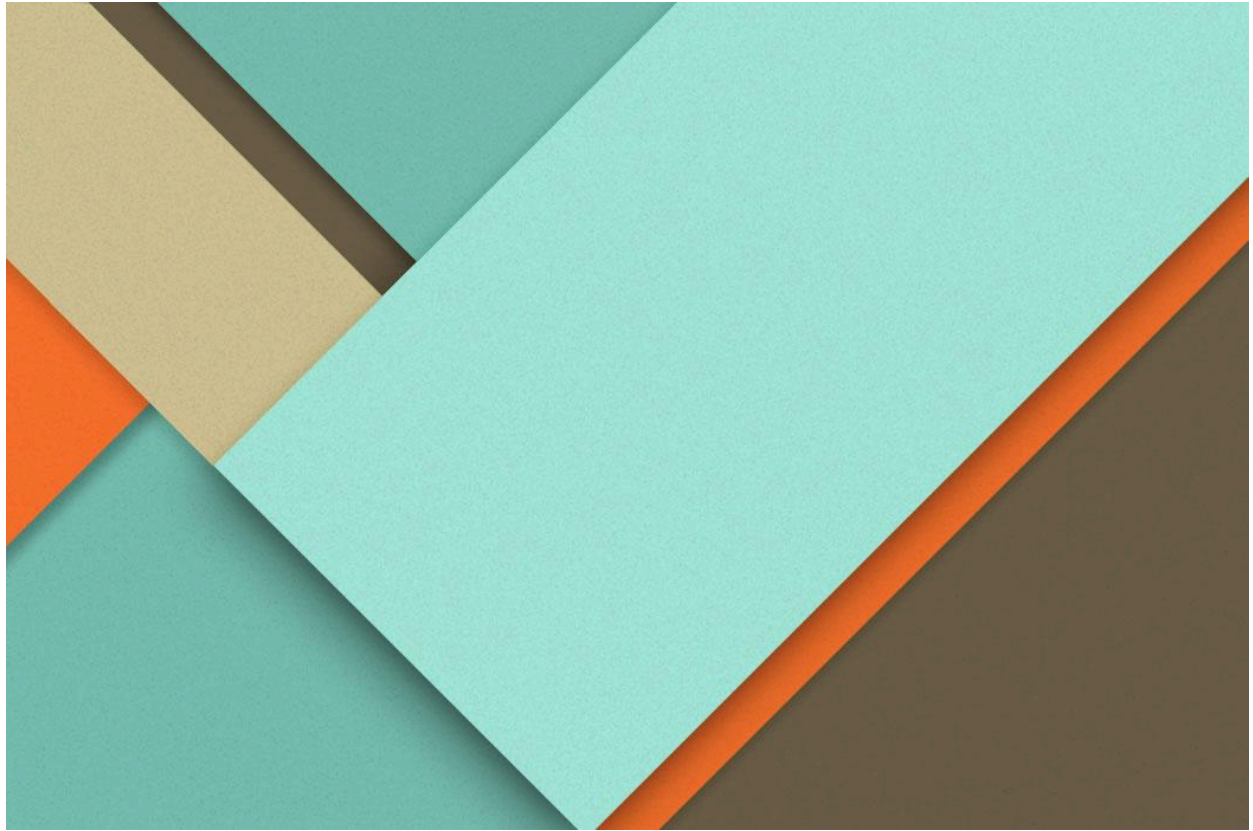




Hospital Management System

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Tutorial: 26

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Overview

This project aims to develop a **Command-Line Interface (CLI) based Hospital Management System** that manages hospital operations such as **patient records, doctor management, appointments, billing, and medical history tracking**.

The system will be implemented using Python with a strong focus on programming concepts such as **object-oriented programming (OOP), loops, conditionals, and functions**. **PostgreSQL** will be integrated to handle structured data storage and retrieval.

All interactions will occur through a terminal interface, ensuring emphasis on backend logic, system design, and database integration.

Goals

1. Build a fully functional CLI-based **hospital management system**
2. Implement database operations using **PostgreSQL**
3. Apply Python concepts (**OOP, loops, conditionals, functions**)
4. Simulate real-world hospital workflows
5. Ensure **efficient** and **structured** data handling

Key Features



Patient Management

- Add new patient records
- Update and delete patient information
- View patient details and medical history



Doctor Management

- Add and manage doctors
- Assign specializations (Cardiology, Neurology, etc.)
- View doctor schedules



Appointment System

- Book appointments between patients and doctors
- Check doctor availability
- Cancel or reschedule appointments



Medical Records

- Store diagnosis and treatment details
- Maintain patient history
- Retrieve past medical records



Billing System

- Generate bills based on consultation and treatment
- Calculate total charges
- Store billing history



Ward / Room Management

- Assign rooms to patients
- Track room availability
- Manage patient admission and discharge



Search & Reports

- Search patients/doctors by ID or name
- Generate reports (appointments, billing, patient data)

Technologies Used

1. **Language:** Python
2. **Database:** PostgreSQL
3. **Library:** psycopg2 (for database connection)
4. **Interface:** Command-Line Interface (CLI)

FlowChart

